

概率论与数理统计练习卷解答

一、填空题（每空 3 分，共 30 分）

1. $\frac{2}{3}$; 2. $(A, \alpha) = (3, 2)$; 3. $b(n_1 + n_2, p)$; 4. $N(2\mu_1 - \mu_2, 4\sigma_1^2 + \sigma_2^2 - 4\rho\sigma_1\sigma_2)$; 5. $P(A)P(B)$;

6. $\Phi\left(\frac{5}{3}\right) = \Phi(1.667) = 0.9525$; 7. $F(n-1, 1)$, 8. $\frac{\sqrt{6}}{2}$; 9. S_1^2 ;

10. $\frac{\sum_{i=1}^n X_i^2}{\chi_\alpha^2(n)}$. 或者 $\frac{(n-1)S^2}{\chi_\alpha^2(n-1)}$ 或者 $\frac{\sum_{i=1}^n X_i^2}{\chi_\alpha^2(n-1)}$ 都正确

二、解答题

11、（10 分）解 记 $A = \{\text{取到的产品是次品}\}$, $B_i = \{\text{取到的产品是 } i \text{ 车间生产的}\}$, $i = 1, 2, 3$. 由全概率公式

$$P(A) = \sum_{i=1}^3 P(A|B_i)P(B_i) = 0.02 \times 0.15 + 0.01 \times 0.80 + 0.03 \times 0.05 = 0.0125 \quad \dots\dots 5 \text{ 分}$$

由 Bayes 公式

$$P(B_1|A) = \frac{P(A|B_1)P(B_1)}{P(A)} = \frac{0.02 \times 0.15}{0.0125} = 0.24$$

$$P(B_2|A) = \frac{P(A|B_2)P(B_2)}{P(A)} = \frac{0.01 \times 0.80}{0.0125} = 0.64$$

$$P(B_3|A) = \frac{P(A|B_3)P(B_3)}{P(A)} = \frac{0.03 \times 0.05}{0.0125} = 0.12$$

由此可见该次品是二车间生产的可能性最大.

$\dots\dots 10 \text{ 分}$

12、(10 分) 解 (1)

$$F_Y(y) = P(Y \leq y) = P\{-2\ln(1-X) \leq y\} = P\left\{\ln(1-X) \geq -\frac{y}{2}\right\}$$

当 $y \leq 0$ 时, $F_Y(y) = 0$; 当 $y > 0$ 时

$$F_Y(y) = P\left\{\ln(1-X) \geq -\frac{y}{2}\right\} = P\left\{X \leq 1 - e^{-\frac{y}{2}}\right\} = 1 - e^{-\frac{y}{2}}$$

故

$$f_Y(y) = \begin{cases} \frac{1}{2}e^{-\frac{y}{2}}, & y > 0 \\ 0, & y \leq 0 \end{cases} \quad \dots\dots 6 \text{ 分}$$

$$(2) \quad f_N(x) = n[1 - F_Y(x)]^{n-1} f_Y(x) = \begin{cases} \frac{n}{2}e^{-\frac{n}{2}x}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad \dots\dots 10 \text{ 分}$$

13、(10 分) 解

$$P\{X = n, Y = m\} = P\{Y = m | X = n\}P\{X = n\} = C_n^m p^m (1-p)^{n-m} \frac{\lambda^n}{n!} e^{-\lambda}$$

$$= \frac{\lambda^n p^m (1-p)^{n-m}}{m!(n-m)!} e^{-\lambda}, \quad m = 0, 1, \dots, n; \quad n = 0, 1, 2, \dots \quad \dots\dots 5 \text{ 分}$$

$$P\{Y = m\} = \sum_{n=m}^{\infty} \frac{\lambda^n p^m (1-p)^{n-m}}{m!(n-m)!} e^{-\lambda} = \frac{(\lambda p)^m}{m!} e^{-\lambda} \sum_{n=m}^{\infty} \frac{[\lambda(1-p)]^{n-m}}{(n-m)!}$$

$$= \frac{(\lambda p)^m}{m!} e^{-\lambda} e^{\lambda(1-p)} = \frac{(\lambda p)^m}{m!} e^{-\lambda p}, \quad m = 0, 1, 2, \dots \quad \dots\dots 10 \text{ 分}$$

即 $Y \sim P(\lambda p)$.

14、(10 分) 解

$$EX = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xf(x, y) dx dy = \int_0^1 x dx \int_{-x}^x 1 dy = 2 \int_0^1 x^2 dx = \frac{2}{3} \quad \dots\dots 2 \text{ 分}$$

$$EX^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 f(x, y) dx dy = \int_0^1 x^2 dx \int_{-x}^x 1 dy = 2 \int_0^1 x^3 dx = \frac{1}{2}$$

$$Var(X) = EX^2 - (EX)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18} \quad \dots\dots 4 \text{ 分}$$

$$EY = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} yf(x, y) dx dy = \int_0^1 dx \int_{-x}^x y dy = 0 \quad \dots\dots 6 \text{ 分}$$

$$E(XY) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyf(x, y) dx dy = \int_0^1 x dx \int_{-x}^x y dy = 0. \quad \dots\dots 8 \text{ 分}$$

$$Cov(X, Y) = E(XY) - EXEY = 0 \quad \dots\dots 10 \text{ 分}$$

15、(15 分) 解：(1)

$$EX = \int_{-\infty}^{\infty} xf(x; \theta) dx = \int_0^1 x \theta x^{\theta-1} dx = \theta \int_0^1 x^{\theta} dx = \frac{\theta}{\theta+1}$$

由矩估计法，令 $\bar{X} = \frac{\theta}{\theta+1}$ ，解得 $\hat{\theta}_M = \frac{\bar{X}}{1-\bar{X}}$. \dots\dots 7 \text{ 分}

(2) 似然函数为

$$L(\theta) = \prod_{i=1}^n \theta X_i^{\theta-1} = \theta^n [\prod_{i=1}^n X_i]^{\theta-1}, \quad \ln L(\theta) = n \ln \theta + (\theta-1) \sum_{i=1}^n \ln X_i$$

$$\frac{d \ln L(\theta)}{d \theta} = \frac{n}{\theta} + \sum_{i=1}^n \ln X_i \triangleq 0$$

解得 $\hat{\theta}_L = -\frac{n}{\sum_{i=1}^n \ln X_i}$, 15 分

16、(15 分) 解 设 X 表示喜欢参加体育运动男生的身高, Y 表示不喜欢参加体育运动男生的身高, 则有

$$X \sim N(\mu_1, \sigma^2), \quad Y \sim N(\mu_2, \sigma^2)$$

我们要做的是对假设

$$H_0: \mu_1 \leq \mu_2 \quad \text{vs} \quad H_1: \mu_1 > \mu_2 \quad \dots\dots\dots 4 \text{ 分}$$

作显著性检验. 在显著性水平 α 下, H_0 的拒绝域为

$$T = \frac{\bar{X} - \bar{Y}}{S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \geq t_{1-\alpha}(n_1 + n_2 - 2) \quad \dots\dots\dots 8 \text{ 分}$$

这里 $n_1 = n_2 = 50$, $\bar{X} = 174.45$, $\bar{Y} = 172.42$, $S_1^2 = (5.35)^2$, $S_2^2 = (6.11)^2$

$$S_w = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{S_1^2 + S_2^2}{2}} = \sqrt{\frac{5.35^2 + 6.11^2}{2}} \approx 5.74$$

$$T = \frac{\bar{X} - \bar{Y}}{S_w \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{174.45 - 172.42}{5.74 \sqrt{\frac{1}{50} + \frac{1}{50}}} = \frac{2.03 \times 5}{5.74} = 1.7683$$

$$t_{1-\alpha}(n_1 + n_2 - 2) = t_{0.95}(98) \approx u_{0.95} = 1.645$$

因 $T = 1.7683 > 1.645$, 故拒绝 H_0 , 即认为喜欢参加体育运动的男生身高比不喜欢参加体育运动的男生身高要高些. \dots\dots\dots 15 分